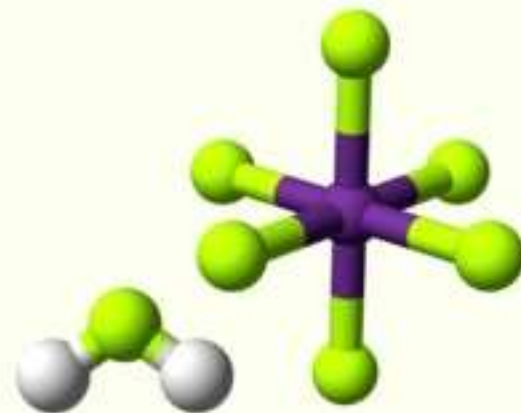


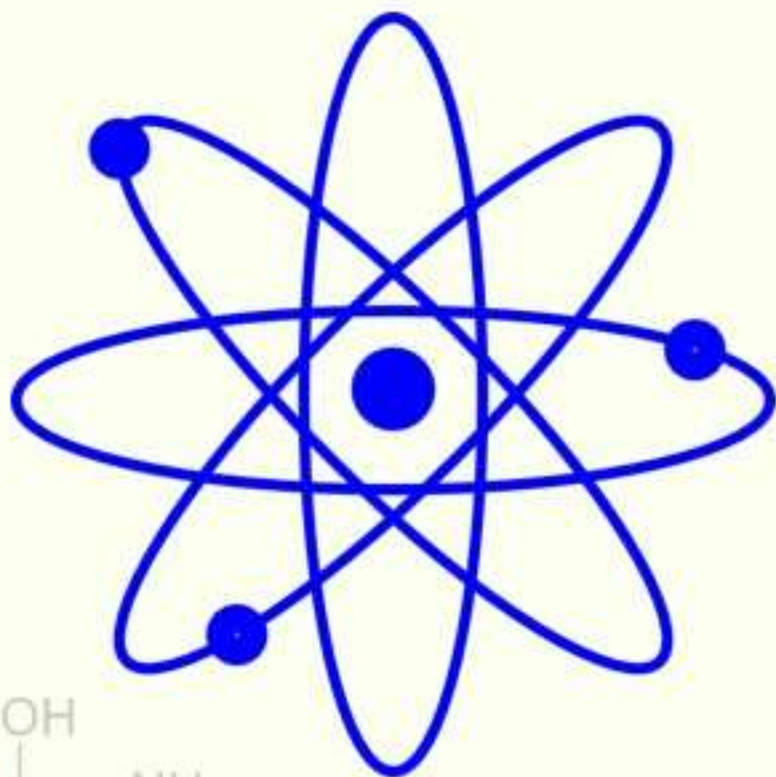
CHEMISTRY

CHEMICAL BONDS

MODULE - 11



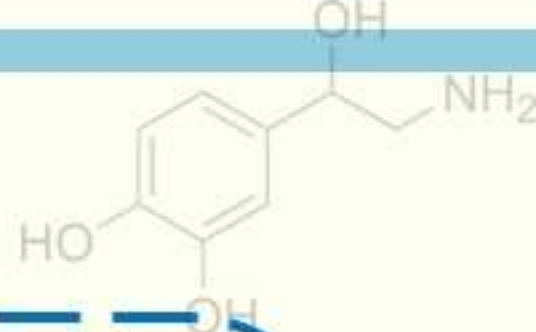
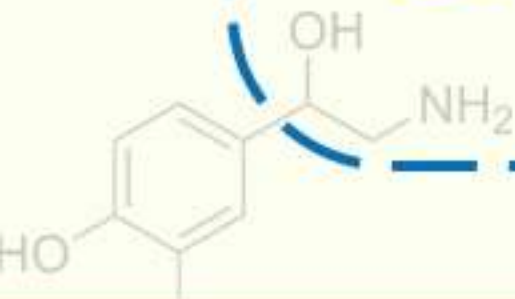
WHY DO CHEMICALS REACT?

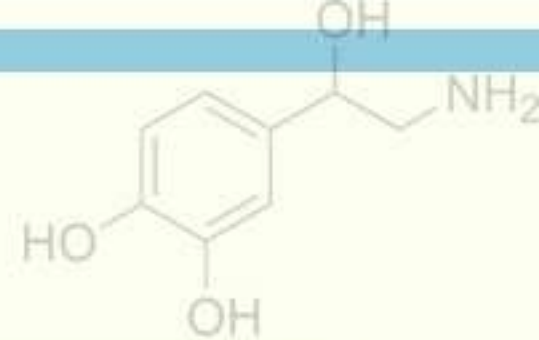


Chemicals react because the molecule break old bonds and make new bonds.

Molecules are formed by the bonds between ions.

Ions are formed when atoms wants to attain the octet configuration.





CHEMICAL BONDS

INTRAMOLECULAR BONDING

*Force of attraction within
the molecule*
Determine Chemical
properties

Ionic bonds

Covalent bonds

Nonpolar

Polar

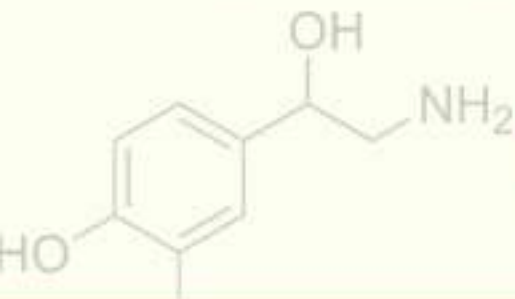
INTERMOLECULAR BONDING

*Force of attraction between
two or more molecules*
Determine physical
properties

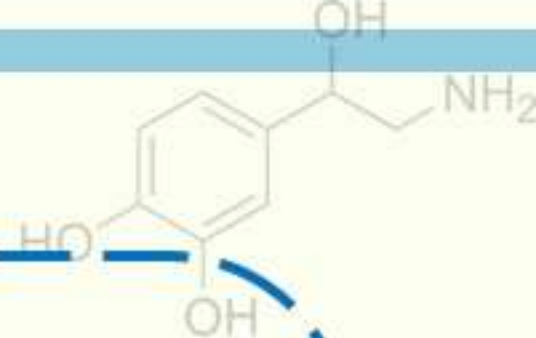
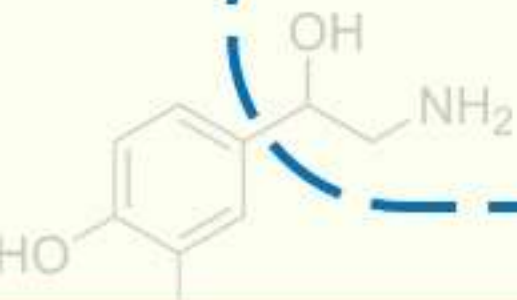
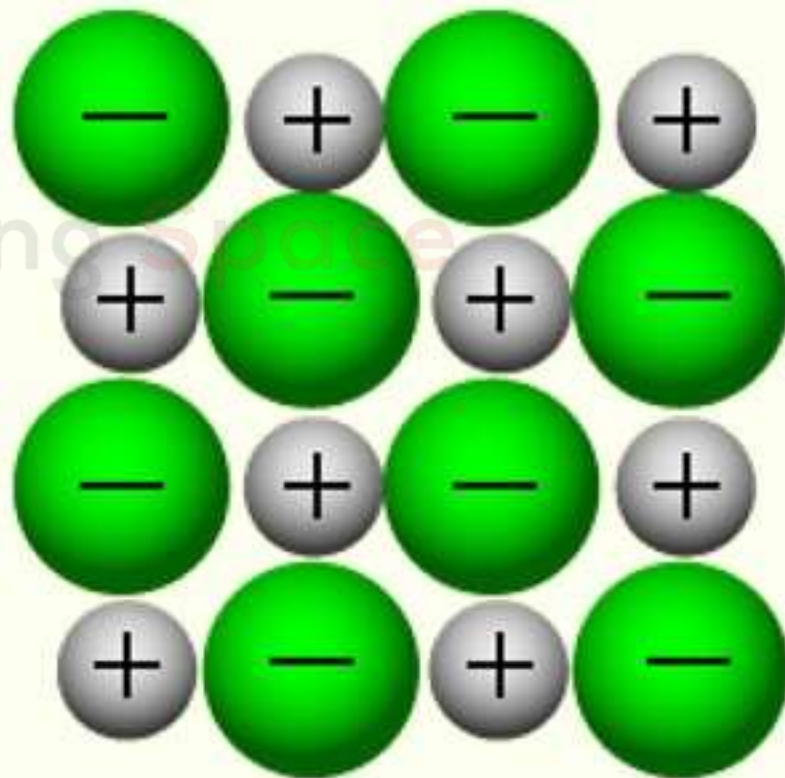
Hydrogen Bonding

Dipole-Dipole interactions

Van der Waals forces

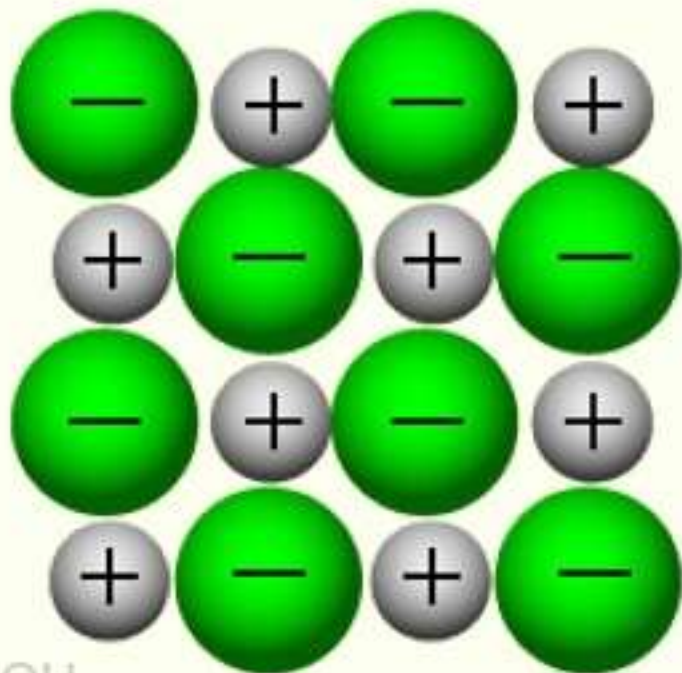


IONIC BOND

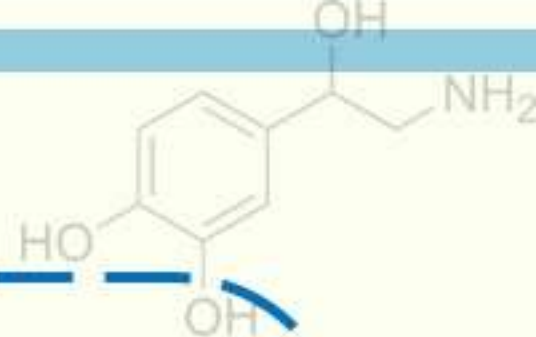
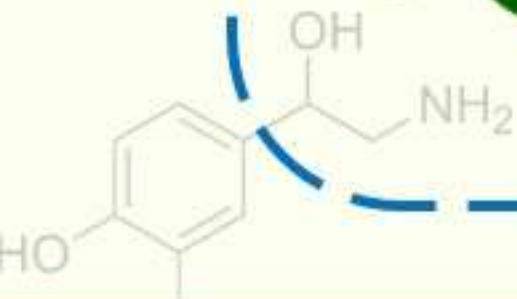




IONIC/ELECTROVALENT BOND

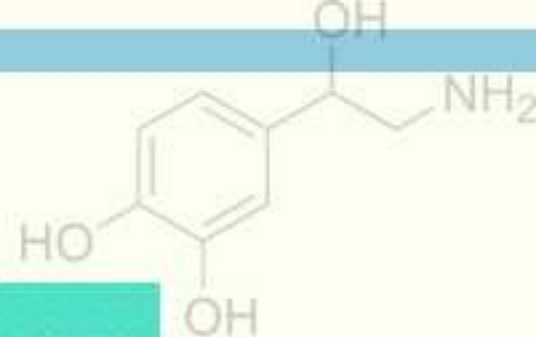


- *The bond formed by the transfer of electrons from one atom to another is called electrovalent bond and the compound is called electrovalent compound or ionic compound.*
- *These bonds are formed between metals and non-metals.*
- *These conduct electricity when dissolved in water and also soluble in water.*
- *These are insoluble in organic solvents like alcohol, acetone*

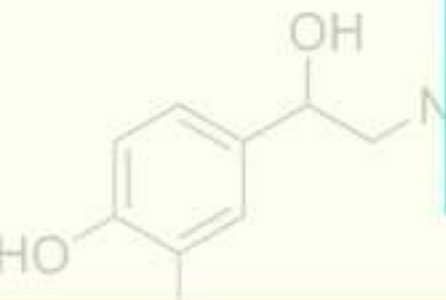




IONIC/ELECTROVALENT BOND

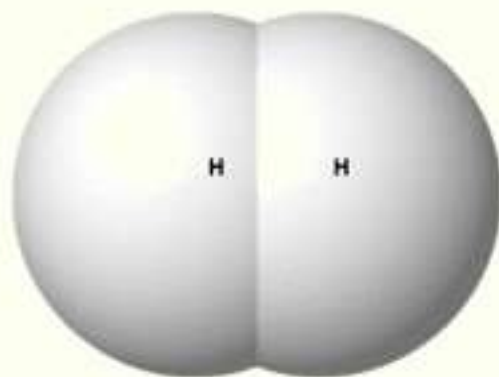


s-Block																		p-Block					
H																		He					
Li	Be											B	C	N	O	F	Ne						
Na	Mg											Al	Si	P	S	Cl	Ar						
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr						
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe						
Cs	Ba	La*	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn						
Fr	Ra	Ac**	Rf	Db	Sg	Bh	Hs	Mt	Uun	Uuu	Uub	Uuq											
f-Block																							
*	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu									
**	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr									





WHICH BOND IS PRESENT IN THE MOLECULE?



**Hydrogen
molecule**



**Oxygen
molecule**

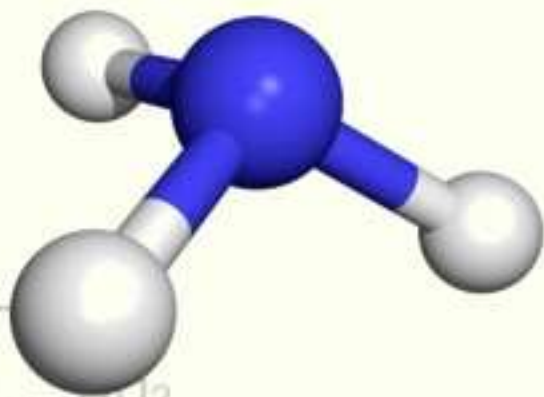
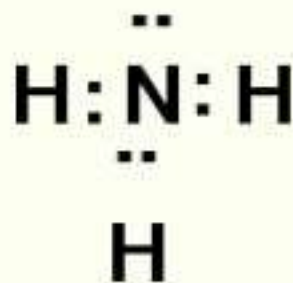


**Water
molecule**





COVALENT BOND

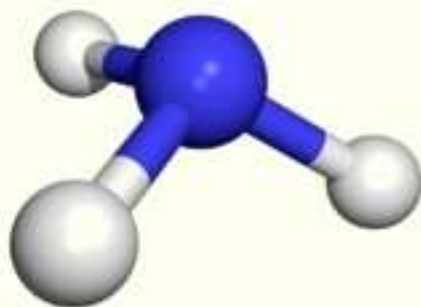
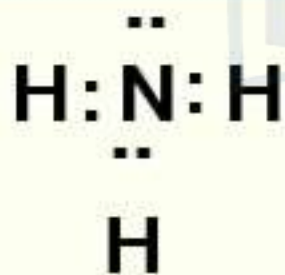


- *The bond is formed by the sharing of electrons between two atoms of same or different elements, is called covalent bond.*
- *Covalent bond may be single, double or triple depends upon the number of sharing pairs of electrons.*
- *Covalent compounds are usually liquids or gases having low melting point and boiling point.*
- *These do not conduct electricity and are insoluble in water.*
- *They can dissolve in organic solvents like alcohol etc.*

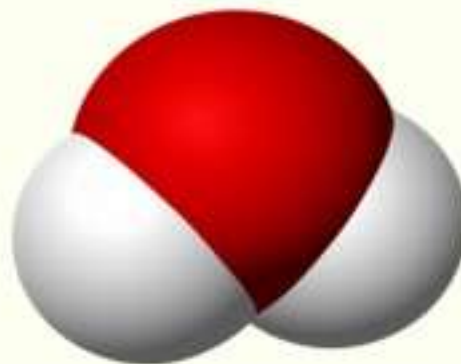


COVALENT BONDS

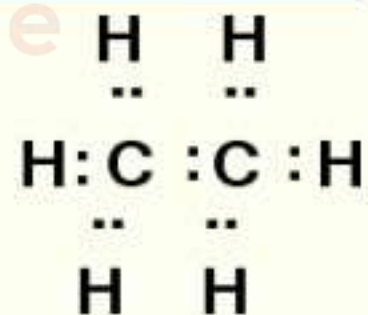
AMMONIA



WATER



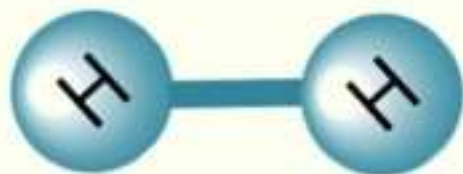
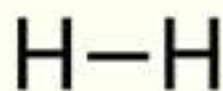
ETHANE



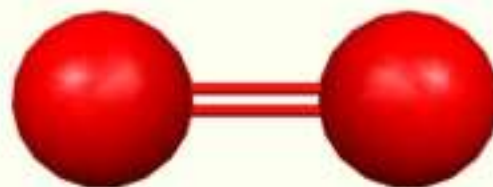
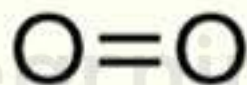


COVALENT BONDS

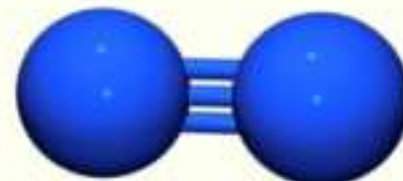
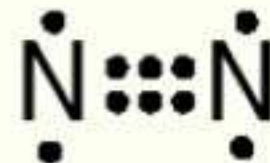
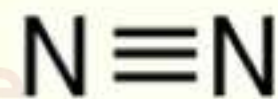
SINGLE BOND



DOUBLE BOND



TRIPLE BOND





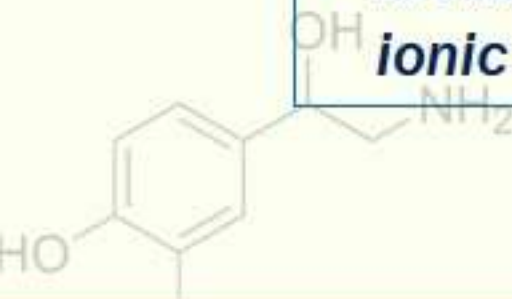
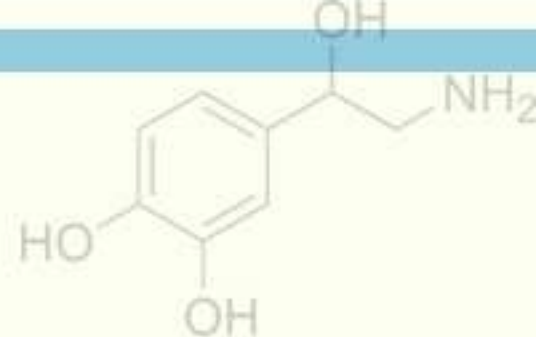
CONTRASTING FEATURES

IONIC BONDS

- *In ionic bonds, electron pairs are shared between atoms.*
- *Ionic bonds occur through the interaction between cations and anions.*
- *Ionic bonds are the strongest type of chemical bond.*
- *Metallic elements tend to form ionic bonds.*

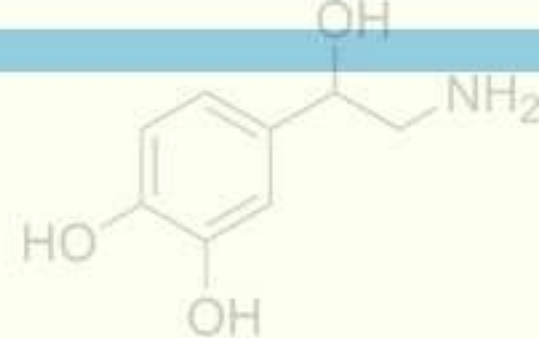
COVALENT BONDS

- *In covalent bonds, atoms are electrostatically attracted towards each other.*
- *Covalent bonds occur through the interaction of neutral atoms.*
- *Covalent bonds are quite weak.*
- *Non metallic elements tend to form covalent bonds.*





COVALENT BONDS



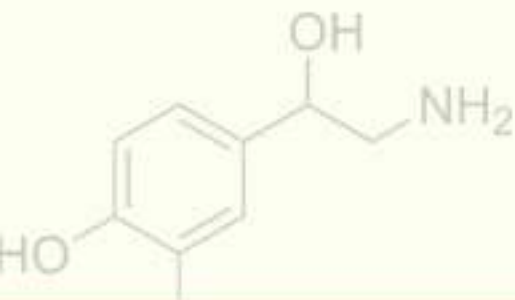
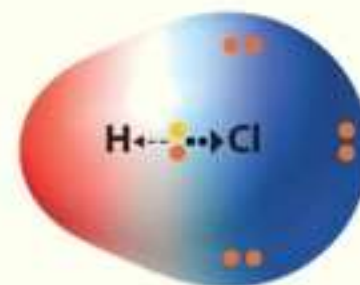
NON-POLAR

- Bonding between electrons **equally** shared between two atoms.
- **No charges** on atoms
- The electronegativity difference is low

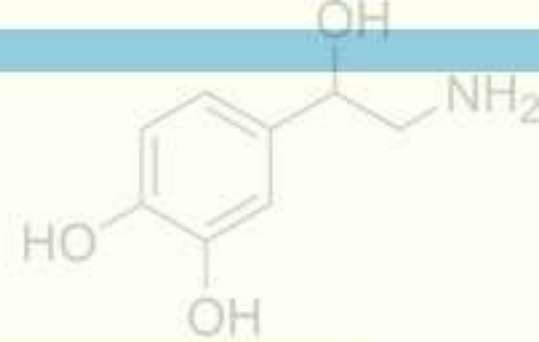


POLAR

- Bonding Electrons shared **unequally** between two atoms.
- **Partial charges** on atoms
- The electronegativity difference is high



CHEMICAL BONDS



INTRAMOLECULAR BONDING

*Force of attraction within
the molecule*
Determine Chemical
properties

Ionic bonds

Covalent bonds

Nonpolar

Polar

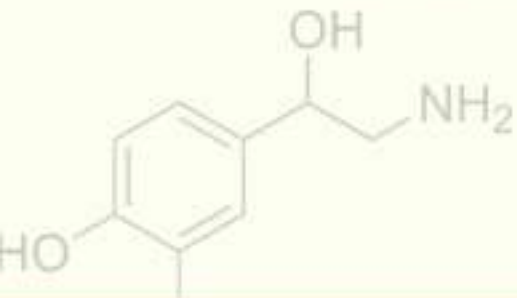
INTERMOLECULAR BONDING

*Force of attraction between
two or more molecules*
Determine physical
properties

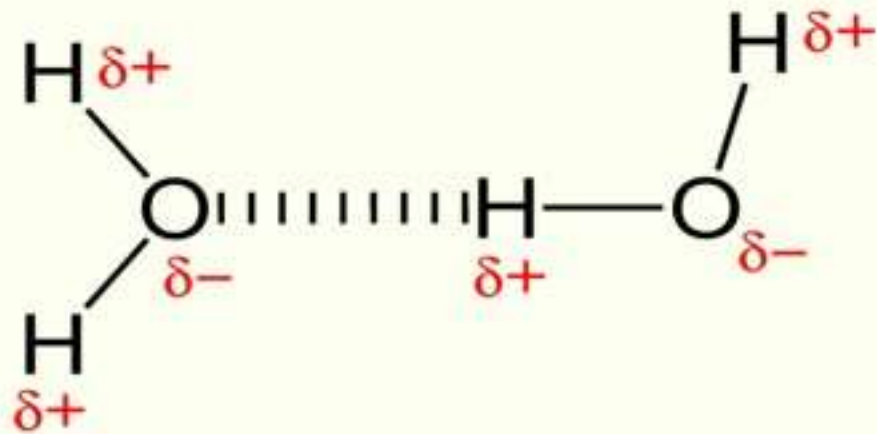
Hydrogen Bonding

Dipole-Dipole interactions

Van der Waals forces

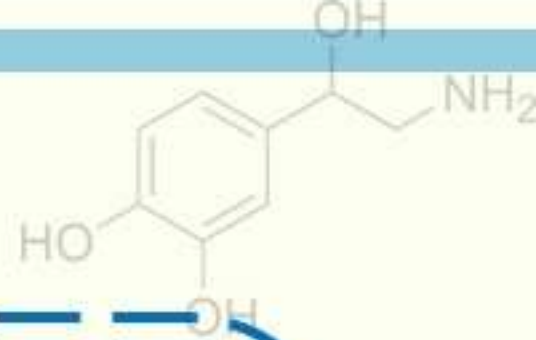
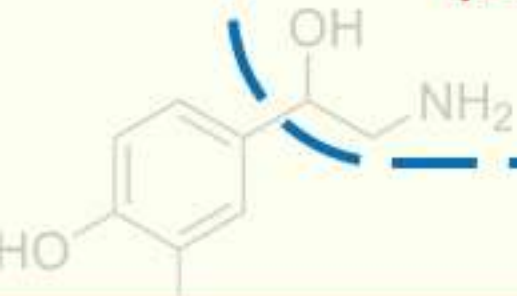


HYDROGEN BOND



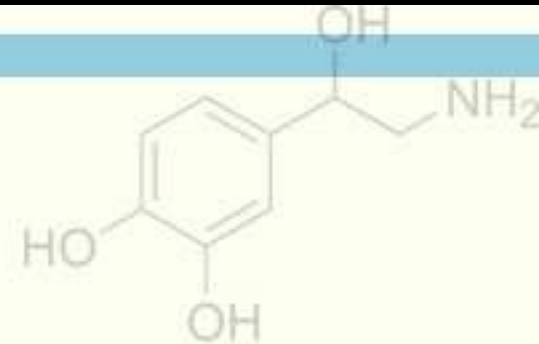
Delta sign represents slightly positive/negative charge.

The electrostatic force of attraction between hydrogen atom and any other electronegative atom which is present in the same or different molecules, is known as hydrogen bond.





HYDROGEN BOND

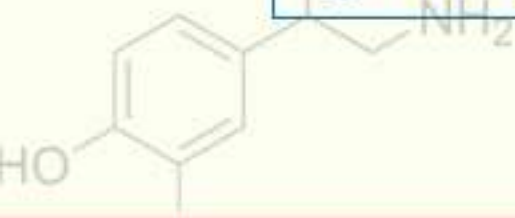


INTRAMOLECULAR HYDROGEN BOND

- Intramolecular hydrogen bonding occurs within the molecule itself
- **Has low melting and boiling points** and high vapour pressure.
- Stability is comparatively low.
Eg. Nitrophenol and salicylic acid.

INTERMOLECULAR HYDROGEN BOND

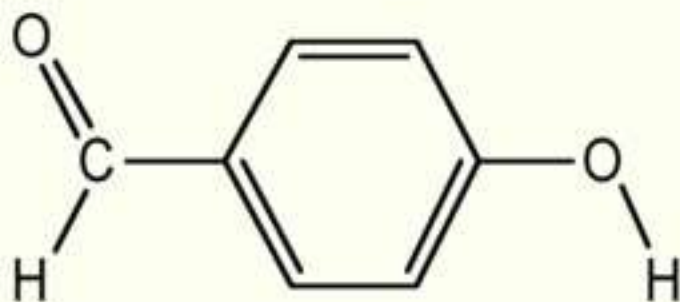
- Intermolecular hydrogen bonding occurs between two neighbouring molecules
- **Has high melting and boiling points**, and low vapour pressure.
- Stability is comparatively high.
Eg. Water, methyl alcohol, ethyl alcohol and sugar



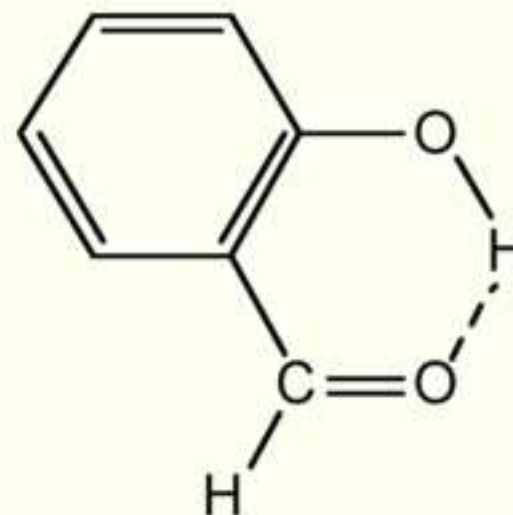


INTRAMOLECULAR HYDROGEN BOND

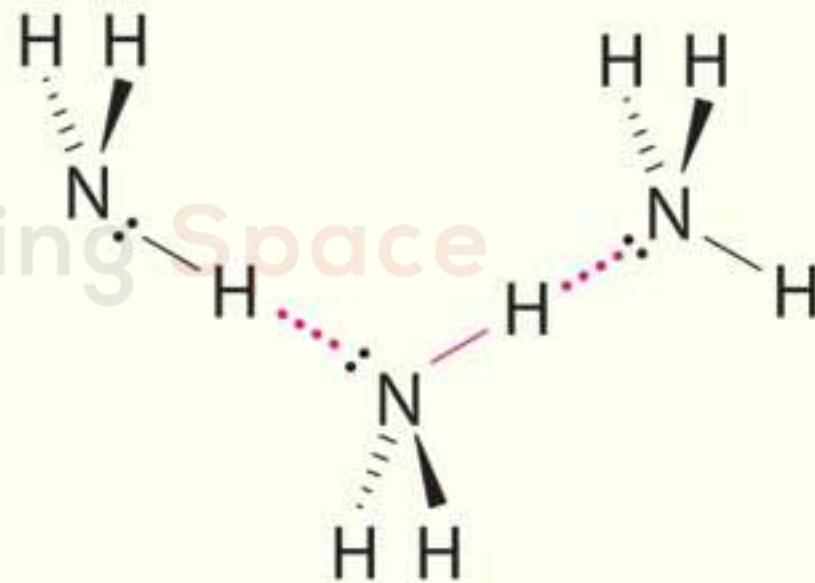
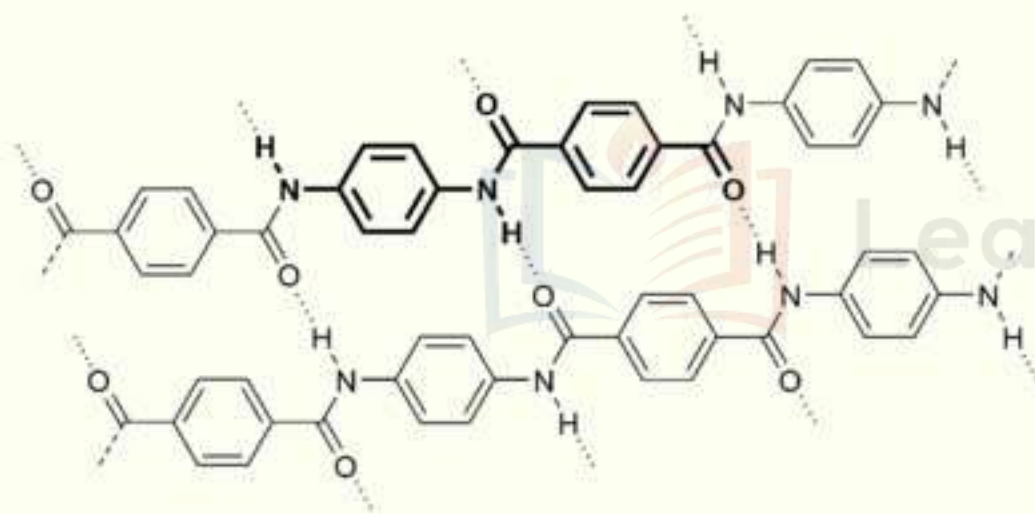
WITHOUT
HYDROGEN BONDING



WITH
HYDROGEN BONDING

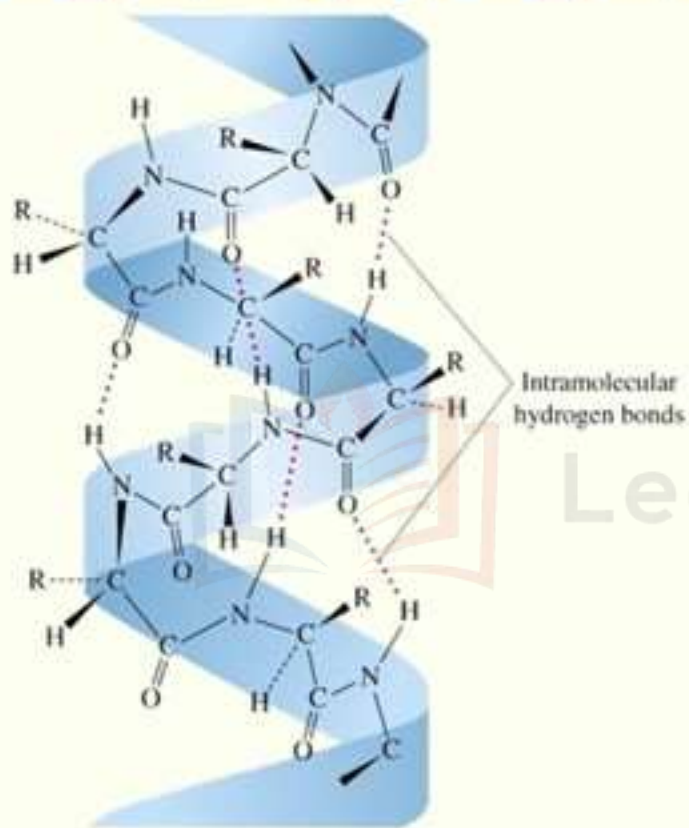


INTERMOLECULAR HYDROGEN BOND





HYDROGEN BONDING IN BIOMOLECULES



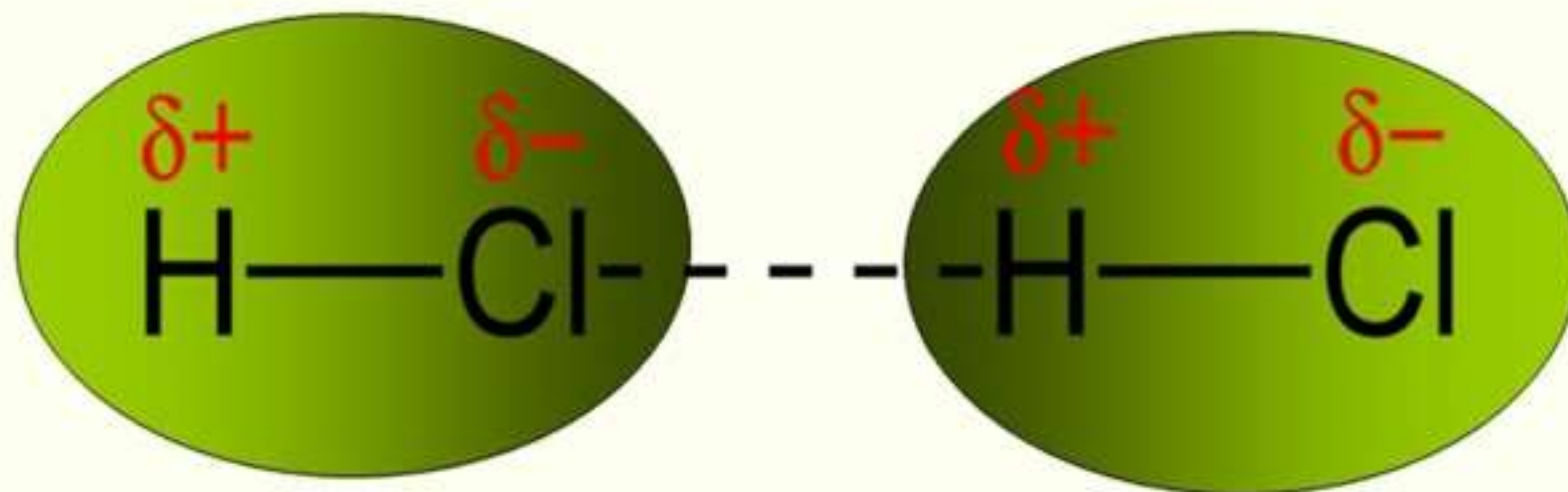
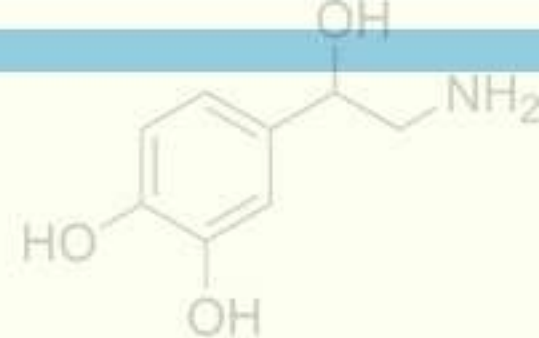
PROTEIN STRUCTURE



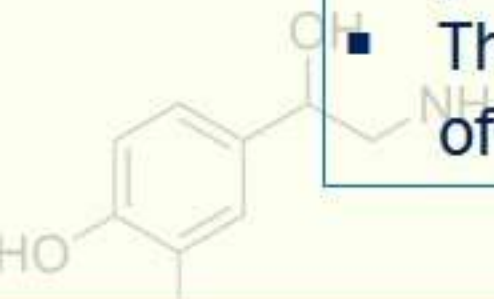
DNA STRUCTURE



DIPOLE-DIPOLE INTERACTIONS

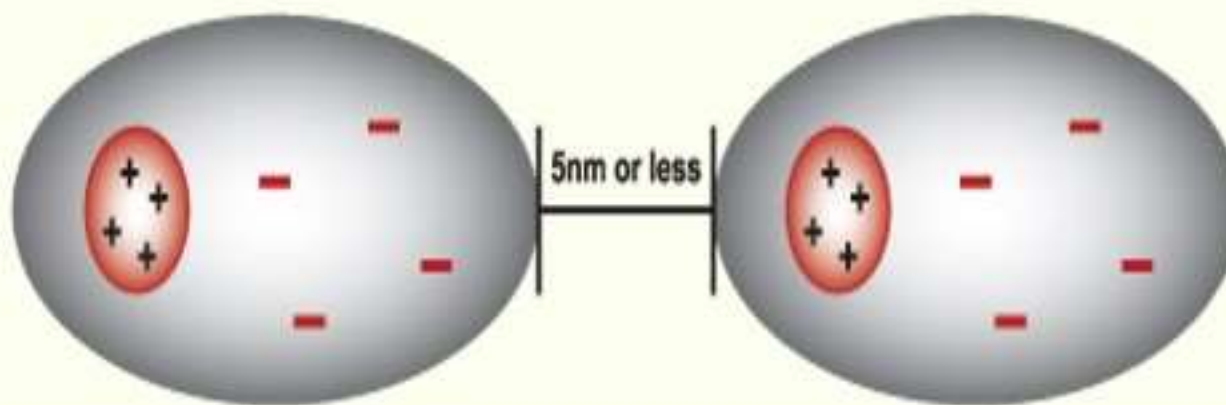
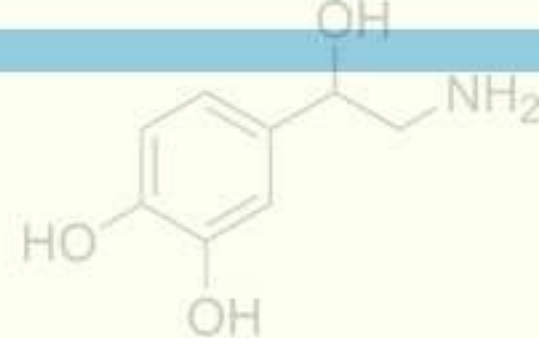


- The forces of attraction between oppositely charged ends of polar molecules.
- The strength of the dipole-dipole is dependent on the polarity of the molecule.

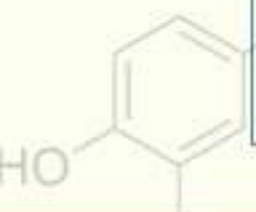




VAN DER WAALS FORCE



- Van der Waals forces or London Forces are **distance-dependent interactions between atoms or molecules**.
- Unlike ionic or covalent bonds, these attractions **are not a result of any chemical electronic bond**.
- Van der Waals forces quickly vanish at longer distances between interacting molecules.
- It loosely symbolizes intermolecular force.



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